ZHENG ZIYUE

+86-15521010105 | zzheng989@connect.hkust-gz.edu.cn | mrwater98.github.io

EDUCATION

PhD | ***Microelectronics*** Sept. 2024 – Present
The Hong Kong University of Science and Technology (Guangzhou)
- Supervised by Yangdi Lyu; Gain Full Scholarship

Redbird MPhil | ***Microelectronics*** Sept. 2022 – Aug. 2024
The Hong Kong University of Science and Technology (Guangzhou)
- Supervised by Yangdi Lyu; Acc GPA: 4.014/4.3 (Full Scholarship)

Bachelor of Science | ***Digital Media Technology*** Oct. 2017 – Aug. 2021
Xiamen University
- Acc GPA: 3.54/4.0

SELECTED PUBLICATIONS

Conference Papers:

GI-RTL-Fuzz: Grammar-Informed RTL Generation with LLM for End-to-End Differential Simulator Fuzzing 2026
Ziyue Zheng, *Zhi Qu, Yangdi Lyu* ICCAD

NeuroAbs: A Neuro-Symbolic RTL Abstraction Framework for Property Checking Acceleration 2026
Zhiyuan Yan, Xiaofeng Zhou, ***Ziyue Zheng****, Ziyi Yang, Wenbin Che, Wei Zhang, Yangdi Lyu, Hongce Zhang* ICCAD

Forbench: Symbolic Simulation Helps Make Your Testbench More Formal 2026
Ziyi Yang, Wenbin Che, ***Ziyue Zheng****, Guangyu Hu, Hongce Zhang* ICCAD

Coverage-Driven RTL Assertion Generation with Formal Exploration and Neuro-Symbolic Refinement 2026
Zhiyuan Yan, ***Ziyue Zheng****, Hongce Zhang* MLCAD

Fine-Grained Code Analysis for Processor Fuzzing 2026
Ziyue Zheng*, Zhi Qu, Yangdi Lyu* DATE

Hot-FV: A Semi-Formal Test Generation Framework for RTL Functional Coverage 2025
Ziyue Zheng*, Zhiyuan Yan, Xiangchen Meng, Guangyu Hu, Hongce Zhang, Yangdi Lyu* ICCD

APE-FV: Concolic Testing for RTL Functional Verification Using Adaptive Path Exploration 2024
Ziyue Zheng*, Xiangchen Meng, Yangdi Lyu* ICCD

FlattenRTL: An Open Source Tool for Flattening Verilog Module at RTL Level 2024
Xiangchen Meng,* ***Ziyue Zheng****, Yangdi Lyu* ISEDA

STSearch: State Tracing-based Search Heuristics for RTL Validation 2023
Ziyue Zheng*, Yangdi Lyu* DATE

Journal Papers:

DeepVerifier: Learning to Update Test Sequences for Coverage-Guided Verification 2025
Yuntao Lu, Chen Bai, Yuxuan Zhao, ***Ziyue Zheng****, Yangdi Lyu, Bei Yu* *TODAES*

State-of-the-Art Advances, Development, and Challenges of Metal Oxide Semiconductor Nanomaterials for Photothermal Solar Steam Generation 2022
Kar Woon Tan, Chun Man Yap, ***Ziyue Zheng****, et al.* *Adv. Sustainable Syst.*

Biosynthesis of Zinc Oxide Nanomaterials from Plant Extracts and Future Green Prospects: A Topical Review 2021
Yajing Huang, Choon Yian Haw, ***Ziyue Zheng****, et al.* *Adv. Sustainable Syst.*

Work Experience

Research Intern Feb. 2026 – Aug. 2026
Huawei Noah's Ark Lab: Multi-modal direction Supervisor: Siyuan XU
- Research & Project: 3DIC Optimization, Multimodal Large Model Full-Stack Optimization.

Research Assistant Sept. 2021 – Sept. 2022
HKUST Fok Ying Tung Research Institute Guangzhou, China
- Research & Project: Optimization Algorithms, Hardware Formal Verification & Testing.

Research Assistant Mar. 2021 – Sept. 2021
Oxford-Hainan Blockchain Research Institution Hainan, China
- Research & Project: Concolic Testing for Smart Contract; Random Number in Blockchain.

Algorithm Engineer Mar. 2020 – Sept. 2020
China Mobile Online Service Company Guangzhou, China
- Research & Project: Deep Semantic Structured Model; English Chatbot through Click-through Data.

Honors and Awards

Third Place EDAthon 2025
Highest Score in Problem from Candence for Hardware Simulator

Bronze Medal 2025 ICT* LLM for Processor Chip Design Competition
CPU Microarchitecture Exploration

Bronze Medal 2025 ICT* LLM for Processor Chip Design Competition
PPA optimization

4th Place 2019 Malaysia National International Collegiate Programming Contest (ACM-ICPC)
Rank: 4/36

Second Prize 2019 China Undergraduate Mathematical Contest in Modeling (CUMCM)
Top 3.26%

Service

HWMCC2024 Benchmark Contributor
https://hwmcc.github.io/2024/

Sub-Reviewer & Student Reviewer
DATE 2024, DAC 2024, HOST 2025, GLSVLSI

Teaching Assistant for C++, Embedded System
2023, 2025, 2026

Huawei "Spark Award" Project Contributor (71st Challenge)
Research on semi-formal modeling for SAT-based hardware verification in EDA topics.

Univista RTL Simulator (UVS) Testing and Verification
Validation and testing of UVS RTL simulator for hardware verification.

ICT* ECOS Team Processor Verification Collaboration
Verification of CL1 (ARM M0-class) core using self-designed coverage-based testing tools.

Skills & Research Interests

Languages: Verilog/SystemVerilog (SVA), Python, C++, Chisel
EDA & Hardware Verification: Symbolic Execution, Concolic Testing, Hardware Simulation & Emulation, LLM-based Testing, SMT Solver, Model Checking, Hardware Security
Optimization & AI: Reinforcement Learning (Multi-agent), Meta-heuristic Optimization, Genetic Algorithm, LLM-driven EDA Optimization
Tools & Frameworks: Cadence JasperGold, Synopsys Formality, Verilator, Yosys, Pono, Cocotb, iverilog, Vcs

*ICT: Institute of Computing Technology, Chinese Academy of Sciences